PBSMS

Tenant AI Assistant — Implementation Plan

Hosted-model route · Milestones M0 to M7 · Version 2.0 · Approved

Item	Value
Document	PBSMS Tenant AI Assistant — Implementation Plan
Version	2.0 — Approved
Status	APPROVED FOR EXECUTION, 31 August 2026. An execution plan, not a requirement: it creates no requirement and amends nothing. Milestones advance by explicit decision, never by drift; see §8.
Implements	Chapter 47 v2.2 (adopted 30 Aug 2026) · Claude as the Tenant AI Assistant v2.0 (adopted 30 Aug 2026)
Constrained by	Internal Engineering Agent v2.1 — EC-005 and EC-507 govern the pbsms-ai/ tree this plan builds inside
Scope	Ask mode only, staff roles, read-only, English, through general availability. Everything else is explicitly deferred; see §2.2.
Provider status	Claude is the presumptive candidate. FR-AIT-708 gates tenant traffic on a three-candidate benchmark at M4.

1. How This Plan Is Shaped

Four constraints determine the sequence. None is negotiable and each one, if ignored, produces work that has to be redone rather than merely delayed.

Constraint	Consequence for sequencing
FR-AIT-708 gates tenant traffic on a three-candidate benchmark	The benchmark needs the harness, the runtime, the retrieval layer, the validator and three adapters. It cannot be early. Everything before it must therefore be built without knowing which model wins — which is achievable, because nothing above the inference contract names a provider.
The evaluation corpus needs domain reviewers, not engineers	It is the longest-lead item in the plan and the one least compressible by adding developers. It starts at M0 and runs continuously. If it slips, M4 slips, and M4 gates everything after it.
Zero-data-retention procurement is per-organisation and eligibility-reviewed	Long lead, outside engineering control, and required before tenant traffic but not before development. Start it at M0 and let it run in parallel.
EC-507 must exist before work begins inside pbsms-ai/	That tree is where M1 starts. The CI check is therefore the first thing built in this plan, ahead of the thing it protects.

1.1 The thin slice

The plan targets one sentence end to end before it targets anything else:

```
A subject teacher asks for their own class's attendance this term
```

```
and receives a validated, cited answer with exact values and
navigable record links — with every other class in the school
unreachable, and the whole interaction in the audit log.
```

That single request exercises authentication, the scope engine, the capability manifest, the orchestrator tool loop, a typed tool under RLS, the evidence envelope, two model calls, all ten validation checks, the renderer, the route resolver, audit and metering. Every zone in the deployables inventory except research. It is the smallest thing that proves the architecture rather than a component of it, and nothing else should be built until it works.

1.2 Explicitly deferred

Deferred	Until
Draft mode	After M6. Introduces the first output that leaves the tenant, and depends on Ask being trusted.
General Conversation	After M6. Adds the FR-AIT-904 ingress path and a new class of stored content.
Research and Controlled Live Internet Research	Chapter 47 §47.18 stage 6. Eight deployables, its own threat model, variable cost.
Combined Analysis	After Research exists. It is the composition of two things that must each work first.
Explain and Find	After M6. Cheap once Ask works; no reason to carry them through the benchmark.
Multilingual	Not authorised until evaluation sets exist for Twi, Ga, Ewe and Hausa.
Parent View	Not authorised by Chapter 47. Separate decision, separate risk review.
Demo mode and billing	M7. Commercial layer once the thing being sold works.

2. Milestones

ID	Objective	Gates on	Exit
`M0`	Decisions, procurement, boundary controls	Nothing	EC-507 green; ZDR in procurement; corpus work started
`M1`	Foundations — contracts, bundle, registry	M0	A release bundle can be built, hashed, registered and rolled back
`M2`	Governance runtime and retrieval	M1	Scope and isolation tests pass with no model in the loop
`M3`	Validator, renderer, thin slice	M2	The §1.1 sentence works end to end against the reference tenant
`M4`	Three adapters and the FR-AIT-708 benchmark	M3 + corpus	Provider selected on measured evidence; reasoning recorded
`M5`	Product integration, audit, metering, cost	M4	All six UI states, cancellation, audit and cost attribution pass
`M6`	Certification and operational shadow	M5 + ZDR	Every blocking gate green on the blind set; shadow review clean

ID	Objective	Gates on	Exit
`M7`	Canary, then general availability	M6	Four consecutive green weeks; block rate ceiling set from measurement

M0 Decisions, Procurement and Boundary Controls

Nothing here is model work. All of it is long-lead or blocking, and most of it is not done by engineers.

Deliverables

Item	Detail	Owner
EC-507 CI check	Fails any Agent pull request touching pbsms-ai/, including rename or move. Built before the tree it protects has anything in it.	Backend
CODEOWNERS and branch protection	Two-approver enforcement on protected zones per EC-202.	Eng Lead
ZDR procurement	Opened with the provider. Per-organisation, eligibility-reviewed, not a toggle. Also opened with the second hosted candidate, because M4 may change the answer.	Privacy/ Legal
Processing register entry	Jurisdiction, lawful basis, carve-out bounds per the amended DP-103. Drafted now, finalised when the agreement is executed.	Privacy/ Legal
Tenant disclosure wording	DP-113 disclosure naming provider, retention, jurisdiction. Needed before canary, drafted now.	Privacy/ Legal
Role and record matrix	Every staff role mapped to modules, records and fields, from Chapter 13. The evaluation corpus cannot be built without it.	Domain + Backend
Corpus programme start	Domain reviewers engaged and the first scenario families defined. This is the long pole; it starts at M0 and never stops.	Evaluation Lead
Threat model	Injection, scope escalation, exfiltration, SSRF (for later), supply chain. Reviewed before M2.	Security

Exit gate

- EC-507 running and failing a deliberately planted violation.
- ZDR requests open with at least two providers.
- Role and record matrix signed off by whoever owns Chapter 13.
- Corpus programme has named reviewers and a first scenario family drafted.

M1 Foundations

The artifacts that make a release identifiable and reversible. Built before any behaviour exists, because retrofitting release identity onto a running system means the first few releases have none.

Item	Detail	Zone
Inference contract	Provider-neutral interface: request, tools, schema, stream, cancel, usage. The narrowest surface that supports the orchestrator. Nothing above it names a provider (NFR-AIT-119).	A

Item	Detail	Zone
constitution.md and policy.yaml	First version. Purpose, capabilities, prohibitions, source rules, refusal rules. Rule IDs matching Chapter 47 requirement IDs so a validator block names the rule it broke.	B
Constitution compiler	Emits system instructions, validator rules and evaluation skeletons. CI fails on a stale derived artifact.	B
response-schema.json	Answer, clarification, refusal, error. No tenant data, no example identifiers, no real-record values (DP-119).	B
tool-catalog skeleton	Schema format and permission tags. One tool implemented at M2.	B
Bundle manifest and registry	Release ID, component versions and hashes, bundle checksum, pinned model identifier, rollback target, deployment history.	B
Behaviour card template	Intended use, prohibited use, provider and snapshot, evaluation results, limitations, failure history.	B
Network zoning	Model zone with no database route. Private data zone with no public egress. Control zone holding secrets.	C

Exit gate

- A bundle can be built, hashed, registered, deployed to a reference environment and rolled back to a prior bundle.
- A constitution change produces a new bundle ID and fails CI if derived artifacts are stale.
- The inference contract compiles with a stub adapter and no provider code anywhere above it.

M2 Governance Runtime and Retrieval

This milestone has no model in it. That is deliberate. Isolation and scope are enforced deterministically, and proving them against a deterministic layer is far cheaper than proving them through a probabilistic one. If a scope bug exists, it should be found here.

Item	Detail	Zone
AI Gateway	Authenticate, rate-limit, open and cancel requests. Client-supplied tenant, school or role identifiers treated as requests to be matched against server state, never as facts.	C
Scope Engine	All eight dimensions from Chapter 47 §47.6.1. Intersection, never union. A broad right in one dimension never overrides a restriction in another.	C
Capability manifest	Server-side, integrity-protected, per request. Only reachable tools and permitted modes. A request with no permitted capability is rejected before inference.	C
Orchestrator	State machine, tool loop, context budget, deadline, retries. No arbitrary model text is ever executed.	C
Session store	Tenant and owner partitioned. Evidence-bound transition per FR-AIT-903, one-way.	C
RLS Tool Gateway	Typed reads on the requester's request-scoped connection. No elevated role, no BYPASSRLS.	D

Item	Detail	Zone
attendance.query	The first and only tool at this milestone. Business-level arguments; the owning module writes the SQL.	D
Evidence envelope	Source class, retrieval timestamp, scope proof, result count, truncation flag, records with render tokens and links.	D
Route resolver	Generates every record link. Model-authored links are stripped, never rendered.	D

Exit gate

- Cross-tenant suite: zero leaks across all three seeded tenants.
- Role suite: every role in the matrix reaches exactly its own records and no others, tested positively and negatively.
- Manifest suite: unauthorised role, disabled tenant, expired subscription and mid-session context switch all produce correct rejections.
- A request with no permitted capability is rejected before any inference call is constructed.

M3 Validator, Renderer and the Thin Slice

The first milestone with a model in the loop, and the first that produces a user-visible answer. The validator is built before the answer is shown, not after it is found to be wrong.

Item	Detail	Zone
Validator — ten ordered checks	Schema, manifest, source mapping, scope, render tokens, prohibited output, links, citations, approval metadata, atomic render. Ordered so the cheapest and most conclusive checks run first.	E
Exact-value renderer	Substitutes verified values into claim templates after validation. The mechanism behind the 100% fidelity gate.	D/E
Block telemetry	Per mode, per role, per bundle. Feeds FR-AIT-706. Built now because a block rate measured from M3 is worth more than one measured at M6.	E/G
Claude adapter	First implementation of the inference contract. Tool handling, streaming, cancellation, usage extraction, error and refusal mapping.	A
Prompt cache boundary	Constitution, tool catalogue and system instructions as a stable prefix; capability manifest, evidence and question after the breakpoint.	A
Thin slice	The §1.1 sentence, end to end, against the seeded reference tenant.	—

Exit gate

- The §1.1 sentence works, repeatedly, with exact values and working links.
- A deliberately corrupted model response — an invented percentage, an out-of-scope record reference, a model-authored link — is blocked by the validator every time, with the rule ID logged.
- Groundedness and numeric fidelity at 100% on the cases that exist so far.
- Block rate recorded and visible, even if the number is currently poor.

M4 Adapters and the FR-AIT-708 Benchmark

This is the decision milestone. Everything before it was built without knowing which model wins. Everything after it depends on the answer, and the answer comes from measurement on PBSMS's own corpus rather than from any assessment document — including the one that proposed Claude.

Item	Detail
Second hosted adapter	A different hosted provider implementing the same inference contract. Also proves NFR-AIT-119 — an abstraction with one implementation is an assumption.
Open-weight adapter	A model PBSMS could realistically self-host. Retained even if self-hosting is not expected to win, because it is the only candidate that answers the snapshot-custody dependency in FR-AIT-701.
Corpus complete for Ask	Golden sets per role, adversarial corpus, blind split by scenario family, dataset hashes recorded.
Scorers and adjudication	Automated scoring per §47.16.2 metric, with ambiguous accuracy, citation and unnecessary-refusal cases routed to two human reviewers and a domain lead on disagreement.
Benchmark run	Identical constitution, tools, evidence and validator. The model is the only variable.
Provider assessment	Each candidate against DP-103, DP-104, DP-107 and DP-113. A candidate failing privacy is excluded regardless of score.
Decision record	Every candidate's scores, the selection, and the reasoning for each rejection. Recorded in the behaviour card and retained.

What to measure

- Every §47.16.2 metric, per candidate, on the blind set.
- Validation block rate per mode and role — the metric most likely to separate candidates, and the one no other gate reveals.
- p95 latency from Ghana to each inference region, against the NFR-AIT-010 budget. Measured, not assumed.
- Cost per request at realistic evidence volumes, with and without prompt caching.

Exit gate

- Three candidates scored on identical inputs.
- Selection made, recorded, and the bundle pinned to the winning model identifier.
- At least two adapters working, one of them the non-selected fallback (NFR-AIT-119).

M5 Product Integration

Item	Detail	Zone
CommandPalette entry	⌘K into the Assistant from any Staff Console screen.	I
Assistant panel	Dockable, renders claims with source-class labels, citations, retrieval timestamp and record links.	I
Six UI states	Loading, success, error, offline, empty, restricted — the restricted state naming what is restricted and who to ask.	I

Item	Detail	Zone
Progress and cancellation	Acknowledgement within the NFR-AIT-010 budget. No substantive content before validation completes.	I
Blocked-response state	An intelligible outcome, counted as a failed request per FR-AIT-707.	I
Admin capability switches	Per-tenant and per-role toggles, effective immediately on live sessions.	I
Feedback control	One action from the response itself, creating a privacy-safe regression obligation.	I/G
Audit writer	Every field in FR-AIT-600 including bundle ID, constitution version, manifest version, validation outcome and cost.	G
Tenant audit viewer	Assistant activity under TEN-022, with private transcript content separated from administrative metadata.	G
Metering and cost ceiling	Accepted requests through the existing TEN-030 and TEN-031 pipeline. Server-side monetary ceiling independent of the request cap.	G

Exit gate

- All six UI states demonstrable, including restricted reached by a stale bookmark rather than a hidden menu item.
- Cancellation stops the in-flight provider request and is not billed to the tenant.
- Cost per request queryable per tenant. Pricing conversations can begin on measured numbers.

M6 Certification and Operational Shadow

Item	Detail
Full blind evaluation	Every gate in §47.16.2 on the held-out set, against the pinned bundle.
Performance and resilience	Latency, throughput, cancellation, timeout, circuit breaker, retry-storm behaviour.
Privacy review	ZDR verified as applied to the specific organisation and workspace. Carve-out bounds present in the executed agreement, not in documentation.
Signed release candidate	Bundle checksum, evaluation report, behaviour card, rollback target.
Operational shadow	Live tenant retrieval under the contractual gates. No output shown to any tenant. Sampled human review of responses.
Incident runbooks	Isolation leak, unverified value, cost anomaly, provider outage, snapshot mismatch, rising block rate.

Exit gate

- Every blocking gate green on the blind set.
- Shadow period complete with sampled review and no blocking incident.
- Rollback tested by actually rolling back, not by asserting it would work.

M7 Canary and General Availability

Item	Detail
Canary cohort	Consenting staff users in a small number of tenants, told plainly the feature is new. Per-role capability flags.
Weekly review	Reported errors, block rate, cost per tenant, latency, privacy events.
Block-rate ceiling	Set from canary measurement per FR-AIT-706. This is the first honest opportunity to set it, and it becomes a promotion gate thereafter.
Pricing decision	§47.11 structure populated with measured cost. Add-on viability confirmed or the feature is repriced.
Demand finding	Whether staff use it and act on its answers. The Chapter 47 §47.18 stage 0.5 question, answered with real users.
General availability	Four consecutive green weeks, stable or falling error and block rates, adoption blockers closed.

M7 is also the first honest opportunity to stop. If canary shows staff do not trust or use the Assistant, the correct outcome is to stop before M8 rather than to expand modes in the hope that more capability produces more use. The governance runtime built to M6 is not wasted in that case — it is required by every future version of this feature, and by the model programme if it is ever revived.

3. Critical Path and Parallelism

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M0 ──┬── EC-507, CODEOWNERS ──────────────┐
     ├── ZDR procurement ──────────────────────────┐
     └── CORPUS PROGRAMME ════════════════════╗    │
                                              ║    │
M1 ── contracts, bundle, registry ──┐         ║    │
                                    │         ║    │
M2 ── scope, manifest, RLS, tools ──┤         ║    │
                                    │         ║    │
M3 ── validator, renderer, slice ───┤         ║    │
                                    │         ║    │
M4 ── adapters ×3, BENCHMARK ═════════════════╝    │
                                    │              │
M5 ── product, audit, metering ─────┤              │
                                    │              │
M6 ── certification, shadow ═══════════════════════╝
                                    │
M7 ── canary, GA ───────────────────┘

══  critical path          ──  ordinary dependency
```

Track	Note
Corpus programme	The critical path. Starts at M0, gates M4, and is limited by domain-reviewer availability rather than engineering capacity. Adding developers does not accelerate it. If anything in this plan is under-resourced, it will be this.
ZDR procurement	Long lead, outside engineering control. Gates M6, not development. Start early and pursue two providers, because M4 may change which one matters.
Frontend	Can begin against a stubbed API from M2 and integrate at M5. The panel, states and citation rendering do not depend on the model decision.

Track	Note
Research (Zone F)	Entirely outside this plan. Do not start it. Eight deployables and a threat model, none of which is needed to answer the demand question at M7.

4. Ownership

Role	Accountable for
Engineering Lead	Milestone advancement decisions, protected-zone approvals, the M4 selection record, rollback authority.
Backend Lead	Inference contract, orchestrator, typed tools, evidence contracts, adapters.
Database/Tenancy Engineer	RLS tool gateway, scope engine, storage partitioning, role fixtures, the M2 isolation gate.
Evaluation Lead	Corpus, scorers, adjudication, CI wiring, the M4 benchmark and its record. The critical path owner.
Security Engineer	Threat model, validator design review, supply chain, network zoning, incident readiness.
Privacy/Legal	ZDR procurement, processing register, carve-out bounds, DP-113 disclosure, jurisdiction analysis.
Domain Reviewers	Role and record matrix, known-correct answers, refusal appropriateness, Ghanaian school terminology.
Product/Design	Modes, source labelling, UI states, feedback, canary cohort and the M7 demand finding.

5. Decision Points

At	Decision	If it goes the other way
M0	Which second hosted and open-weight candidates enter the benchmark	Choosing late compresses M4. Choose at M0 even if provisionally.
M2	Whether the scope engine handles all eight dimensions or a reduced set for v1	A reduced set is acceptable only if the omitted dimensions are structurally unreachable, not merely unused. Record as a deviation if reduced.
M4	Provider selection	If Claude does not win, the presumptive adoption lapses and the bundle pins the winner. No rework above the inference contract.
M4	Whether any candidate meets the gates at all	If none does, Chapter 47 §47.18 stage 3 adaptation becomes live rather than conditional, and this plan is replaced by a longer one.
M6	Whether ZDR and carve-out bounds are contractually acceptable	If not, that provider is excluded regardless of benchmark score, and the fallback adapter carries the release.
M7	Whether the add-on clears its own cost	If not, reprice or withdraw. Do not ship a feature that loses money per query on a per-active-student subscription.
M7	Whether staff use it	If not, stop. Expanding modes to rescue adoption is the

At	Decision	If it goes the other way
		failure pattern this plan is sequenced to avoid.

6. Risks to the Plan

Risk	Signal	Response
Corpus under-resourced	M4 date slipping while engineering milestones hold	The only real fix is more domain-reviewer time. Reducing corpus scope reduces what the benchmark can prove, which defeats its purpose.
Benchmark treated as a formality	One candidate implemented properly and two implemented hastily	The comparison must hold everything but the model identical. A rushed adapter produces a result that looks like a model finding and is an implementation finding.
Block rate discovered late	Not measured until M6	Measure from M3. A poor early number is information; a poor late number is a schedule problem.
Scope creep into Draft or Research	Requests to add modes before M7	Every mode added before the demand question is answered multiplies the cost of a negative answer.
ZDR arrives late	M6 blocked with everything else green	Development is unaffected; only shadow and canary are blocked. Start procurement at M0 and do not treat it as a checkbox.
Latency from Ghana misses budget	Measured at M4	Region selection, prompt caching and a smaller routing model are the levers. Measuring at M4 leaves time to pull them.
Provider snapshot retired mid-plan	Provider deprecation notice	The fallback adapter exists precisely for this. Re-pin, rerun gates, redeploy as a new bundle.

7. Definition of Done

Area	Evidence
Release identity	A pinned bundle with checksum, behaviour card, evaluation report and a rollback target that has been exercised.
Isolation	Cross-tenant and role suites at zero across all three seeded tenants, run on every release.
Grounding	Every tenant claim traceable to Operational Evidence; ungrounded output blocked, not corrected.
Exact values	100% numeric, date, identifier and name fidelity through the renderer.
No writes	No write tool, no write credential, no external-action route anywhere in the tool catalogue.

Area	Evidence
Selection defensibility	Three candidates scored on identical inputs, with rejection reasoning recorded and retained.
Privacy	ZDR verified on the specific organisation; carve-out bounded in the executed agreement; jurisdiction registered and disclosed.
Operations	Cancellation, audit, metering, cost ceilings, alerts, rollback and incident runbooks all tested rather than documented.
Demand	Canary evidence that authorised staff use the Assistant and act on its answers.

The last row is the one that decides whether any of the others mattered. Every technical gate in this plan can be green while the feature is unused. Chapter 47 §47.19 has recorded that as the principal commercial risk since v2.0, and M7 is where it stops being a risk and becomes a finding.

8. Execution Control

The plan is approved for execution as of 31 August 2026. What follows governs how it is run, because a plan without advancement rules becomes a description of what happened rather than a control on what happens.

8.1 Milestone advancement

- A milestone advances by an explicit decision of the Engineering Lead against its stated exit gate, recorded with the date and the evidence relied on. Milestones do not advance by consensus, by calendar, or because the previous one has been running a while without incident.
- A partially met exit gate is not met. Where a gate cannot be met and the work must proceed, that is a named deviation under Chapter 47 §47.0.1 with its reasoning recorded — not a judgement call made in a standup.
- M4 and M7 additionally require the decision record described in their own sections. M4 without a recorded three-candidate comparison does not satisfy FR-AIT-708, and a release that proceeds on it would be non-compliant with an adopted requirement rather than merely undocumented.

8.2 Amending the plan

- Milestone content may be resequenced by the Engineering Lead where dependencies allow, without a version increment. The plan is a working document and over-formalising it makes it stale.
- Three things may not be resequenced without a version increment and re-approval: the position of the FR-AIT-708 benchmark before tenant traffic; the M2 requirement that isolation is proven without a model in the loop; and the deferral list at §1.2.
- Adding a mode to the M1 to M7 scope is a re-approval, not a resequencing. Every mode added before the M7 demand question is answered multiplies the cost of a negative answer, which is the specific failure this sequence exists to prevent.

8.3 Reporting

Cadence	Content
Weekly, from M0	Corpus programme progress against the M4 date. It is the critical path and it is the item most likely to slip quietly, because engineering milestones will continue to look healthy

Cadence	Content
	while it does.
Weekly, from M3	Validation block rate by mode and role, with the trend. Recorded from the first working answer, not from certification.
At each milestone	Exit-gate evidence, decisions taken, deviations recorded, and the revised date for the next gate.
Weekly, from M7	Canary error reports, block rate, cost per tenant, latency and privacy events.

8.4 Starting conditions

M0 is cleared to begin immediately. Three of its items are on the critical path or have external lead time and should start this week rather than after the others are planned:

- EC-507, because M1 opens work inside pbsms-ai/ and the check must exist before the tree it protects holds anything.
- The corpus programme, because it gates M4 and cannot be compressed by adding engineers.
- Zero-data-retention procurement with two providers, because it is outside engineering control and gates M6.

Nothing else in M0 is urgent in the same way. The role matrix, threat model and disclosure wording are all necessary and none of them has a lead time measured in weeks. The three above do, and starting them late is the most likely way this plan slips without anyone noticing until M4.

— End of Plan —